

PRACTO – DOCTOR APPOINTMENT MANAGEMENT SYSTEM

1. Title

Project Name

Practo – Doctor Appointment Management System

2. Introduction

Healthcare services have evolved significantly with the advancement of digital technology. Nowadays, patients prefer booking doctor appointments online instead of visiting hospitals personally or making phone calls. The traditional appointment booking process is time-consuming, difficult to manage, and often results in scheduling conflicts and long waiting times.

The **Practo – Doctor Appointment Management System** is a web-based healthcare application developed using **Python (Django Framework), HTML, CSS, JavaScript, and MySQL**. The primary objective of this system is to simplify the appointment booking process by allowing users to search for doctors, view doctor profiles, and book appointments online from anywhere.

The application provides two main roles:

- **Admin**
- **User**

The Admin manages doctors, users, appointments, and overall system records through an administrative dashboard, while Users can register, log in, search doctors, and book appointments according to their preferred date and time.

The project minimizes paperwork, improves record management, enhances efficiency, and provides a secure and user-friendly healthcare management solution.

3. Objectives

The major objectives of the Practo Doctor Appointment Management System are:

- To provide an online doctor appointment booking system.
- To reduce manual paperwork in hospitals and clinics.
- To simplify appointment scheduling.

- To maintain patient appointment records digitally.
 - To provide a secure login system for users and administrators.
 - To improve doctor management.
 - To reduce waiting time for patients.
 - To improve communication between doctors and patients.
 - To provide a responsive and user-friendly interface.
 - To ensure data security and integrity.
-

4. Project Overview

The Practo Doctor Appointment Management System is a web-based application that enables patients to search for doctors, view their profiles, and book appointments online. The application is designed to replace the traditional appointment booking system with a digital solution that is efficient, reliable, and easy to use.

The project is developed using **Python (Django Framework)** as the backend technology, **HTML, CSS, Bootstrap, and JavaScript** for the frontend, and **MySQL** as the database.

The system contains two major users:

Admin

The Admin is responsible for managing the complete application. The administrator can add, update, and delete doctor records, manage users, view appointments, approve or reject appointments, and monitor overall system activities through a secure dashboard.

User

The User can register an account, log in securely, search doctors by specialization, view doctor profiles, book appointments, manage personal information, and view appointment history.

The system provides an efficient, secure, and paperless healthcare appointment management solution.

5. Project Scope

A. Included Scope

The current version of the project includes the following functionalities:

- User Registration

- User Login
 - Admin Login
 - Secure Authentication
 - Doctor Management
 - User Management
 - Search Doctor
 - Doctor Profile
 - Book Appointment
 - Appointment History
 - Appointment Status
 - Dashboard
 - Profile Management
 - Logout
 - Database Management
-

B. Excluded Scope

The following features are not included in the current version but can be implemented in future versions:

- Online Payment Gateway
- Video Consultation
- Live Chat with Doctors
- Medicine Ordering
- Laboratory Test Booking
- Ambulance Booking
- Email Notifications
- SMS Notifications
- AI-based Doctor Recommendation
- Mobile Application

- Online Prescription Download
 - Multi-Hospital Support
-

6. BRD (Business Requirement Document)

A. Functional Requirements

Admin Functional Requirements

The Admin shall be able to:

- Login securely using valid credentials.
 - Access the Admin Dashboard.
 - Add new doctor records.
 - Update doctor information.
 - Delete doctor records.
 - View all registered users.
 - Manage user accounts.
 - View all appointments.
 - Approve appointments.
 - Reject appointments.
 - Update appointment status.
 - Manage doctor specializations.
 - View reports and statistics.
 - Logout securely.
-

User Functional Requirements

The User shall be able to:

- Register a new account.
- Login securely.
- Update profile information.
- Search doctors.

- View doctor profiles.
 - Filter doctors based on specialization.
 - Book appointments.
 - View appointment history.
 - Cancel appointments before the scheduled date.
 - Receive appointment status updates.
 - Logout securely.
-

B. Non-Functional Requirements

Admin

The system should provide:

- Secure authentication.
 - Fast dashboard loading.
 - Reliable database connectivity.
 - Data security.
 - High availability.
 - Easy navigation.
 - Efficient appointment management.
 - Proper error handling.
 - Database backup support.
-

User

The application should provide:

- User-friendly interface.
- Responsive web design.
- Fast page loading.
- Secure login system.
- Privacy of personal information.

- Easy appointment booking.
 - Reliable performance.
 - Cross-browser compatibility.
 - Data integrity.
 - Simple navigation.
-

7. FRD (Functional Requirement Document)

7.1 Introduction

The Functional Requirement Document (FRD) describes the detailed functionality of the Practo – Doctor Appointment Management System. It explains how each module of the application will work and how users will interact with the system. The FRD acts as a blueprint for the development process by defining the expected behavior of every feature.

The system is divided into two primary roles: **Admin** and **User**. Each role has specific responsibilities and permissions to ensure secure and efficient operation of the application.

7.2 Functional Requirements

A. User Registration Module

Description

The system shall allow new users to create an account by providing their personal details.

Input

- Full Name
- Email Address
- Mobile Number
- Password
- Confirm Password

Process

- Validate user input.

- Check if the email already exists.
- Encrypt the password before storing it.
- Save user details in the database.

Output

- Successful registration message.
 - Redirect user to the login page.
-

B. User Login Module

Description

The system shall authenticate registered users using valid login credentials.

Input

- Email
- Password

Process

- Validate credentials.
- Verify password.
- Create user session.

Output

- Redirect to User Dashboard.
 - Display error message for invalid credentials.
-

C. Doctor Management Module (Admin)

Description

The Admin shall manage doctor information.

Functionalities

- Add Doctor
- Update Doctor
- Delete Doctor

- View Doctor Details
- Manage Doctor Specializations

Output

Updated doctor records displayed successfully.

D. Doctor Search Module (User)

Description

Users can search doctors based on specialization or doctor name.

Functionalities

- Search by Name
- Search by Department
- View Doctor Profile

Output

List of matching doctors.

E. Appointment Booking Module

Description

Users can book appointments with available doctors.

Input

- Doctor Selection
- Appointment Date
- Appointment Time
- Patient Remarks (Optional)

Process

- Verify doctor's availability.
- Store appointment information.
- Set appointment status to **Pending**.

Output

Appointment booked successfully.

F. Appointment Management Module (Admin)

Description

The Admin manages all appointment requests.

Functionalities

- View Appointments
 - Approve Appointment
 - Reject Appointment
 - Cancel Appointment
 - Update Appointment Status
-

G. User Profile Module

Description

Users can manage their personal information.

Functionalities

- Edit Profile
 - Change Password
 - Update Contact Details
 - View Profile Information
-

H. Dashboard Module

Admin Dashboard

The dashboard shall display:

- Total Doctors
- Total Users
- Total Appointments
- Pending Appointments

- Approved Appointments
- Cancelled Appointments

User Dashboard

The dashboard shall display:

- Upcoming Appointments
 - Appointment History
 - Doctor Recommendations (Future Enhancement)
 - Profile Summary
-

7.3 Non-Functional Requirements

Performance

- The application should load pages quickly.
 - The system should efficiently handle multiple users simultaneously.
-

Security

- Passwords must be encrypted.
 - Only authenticated users can access their dashboards.
 - Unauthorized access should be restricted.
 - Session management should be implemented.
-

Reliability

- The application should remain available during normal operating hours.
 - Data should be stored securely without loss.
-

Availability

- The system should be accessible 24×7 through the internet.
-

Scalability

The application should support future expansion such as:

- Multiple Hospitals
 - Multiple Clinics
 - Online Consultation
 - Payment Gateway
-

Maintainability

The application should be easy to update and maintain.

Usability

The interface should be simple, clean, responsive, and easy for users of all age groups.

Compatibility

The application should work properly on:

- Google Chrome
 - Microsoft Edge
 - Mozilla Firefox
 - Mobile Browsers
-

Data Integrity

The system should ensure that all records remain accurate, complete, and consistent.

8. Business Rules

The following business rules are applied in the Practo – Doctor Appointment Management System:

1. Every user must register before booking an appointment.
2. Email addresses must be unique.
3. Only registered users can log in.

4. Only the Admin can manage doctor records.
 5. Every doctor must belong to a specialization.
 6. Users can only book appointments with available doctors.
 7. One appointment belongs to one registered user.
 8. Appointment status can only be changed by the Admin.
 9. Users can cancel only their own appointments.
 10. Admin has complete access to the system.
 11. Unauthorized users cannot access the Admin Dashboard.
 12. Every appointment must have a valid date and time.
 13. Invalid login credentials should display an error message.
 14. Passwords should always be stored in encrypted form.
 15. User information should remain confidential.
-

9. Comparing Brands

Feature	Practo	Project Traditional Hospital System
Online Appointment Booking	Yes	No
User Registration	Yes	No
Doctor Search	Yes	Limited
Digital Records	Yes	No
Appointment History	Yes	No
Secure Login	Yes	No
Doctor Profile	Yes	No
Admin Dashboard	Yes	No
Paperless Process	Yes	No
Appointment Status Tracking	Yes	No
Easy Management	Yes	No

Feature	Practo Project Traditional Hospital System	
---------	--	--

Time Saving	Yes	No
-------------	-----	----

The proposed Practo system offers a modern, efficient, and user-friendly solution compared to traditional manual appointment systems. It improves patient convenience, reduces paperwork, and enables better management of healthcare services.

10. Technology Stack

The Practo – Doctor Appointment Management System is developed using modern web technologies that provide security, scalability, and a user-friendly experience.

10.1 Frontend

The frontend is responsible for the visual appearance and user interaction of the application.

Technologies Used

- HTML5
- CSS3
- Bootstrap 5
- JavaScript

Purpose

- Design responsive web pages.
 - Create attractive user interfaces.
 - Perform client-side validation.
 - Improve user experience.
-

10.2 Backend

The backend handles all business logic and application processing.

Technology Used

- Python
- Django Framework

Purpose

- User Authentication
 - Appointment Management
 - Doctor Management
 - User Management
 - Database Operations
 - Session Management
 - Security
-

10.3 Database

Technology Used

- MySQL

Purpose

The database stores all application data securely.

It stores:

- User Details
 - Doctor Information
 - Appointment Records
 - Login Credentials
 - Specializations
-

10.4 Deployment

The application is developed locally using:

- Django Development Server
- Visual Studio Code
- MySQL Server
- Windows Operating System

Future deployment platforms include:

- PythonAnywhere

- Render
 - Railway
 - AWS (Future)
-

10.5 Version Control System

The project uses:

- Git
- GitHub

Benefits

- Source Code Management
 - Version Tracking
 - Team Collaboration
 - Backup of Project
-

11. System Modules

The Practo Doctor Appointment Management System consists of the following modules.

11.1 Authentication Module

Features:

- User Registration
 - User Login
 - Admin Login
 - Logout
 - Password Security
-

11.2 User Module

The User Module provides the following functionalities:

- Register Account

- Login
 - View Doctors
 - Search Doctors
 - Book Appointment
 - View Appointment History
 - Cancel Appointment
 - Update Profile
-

11.3 Admin Module

The Admin Module provides complete administrative control over the system.

Features:

- Dashboard
 - Manage Doctors
 - Manage Users
 - Manage Appointments
 - Manage Specializations
 - View Reports
-

11.4 Doctor Module

This module manages doctor information.

Features:

- Add Doctor
 - Update Doctor
 - Delete Doctor
 - View Doctor Details
 - Doctor Availability
-

11.5 Appointment Module

Responsible for appointment scheduling.

Features:

- Book Appointment
 - Cancel Appointment
 - Approve Appointment
 - Reject Appointment
 - Appointment History
-

11.6 Profile Module

Features:

- Edit Profile
 - Update Personal Information
 - Change Password
-

12. Database Overview

The application uses MySQL as the backend database.

The proposed database consists of the following tables:

User Table

Stores:

- User ID
 - Name
 - Email
 - Phone Number
 - Password
-

Doctor Table

Stores:

- Doctor ID

- Doctor Name
 - Specialization
 - Qualification
 - Experience
 - Consultation Fee
 - Availability
-

Appointment Table

Stores:

- Appointment ID
 - User ID
 - Doctor ID
 - Appointment Date
 - Appointment Time
 - Appointment Status
-

Specialization Table

Stores:

- Specialization ID
 - Specialization Name
-

13. Advantages

The Practo Doctor Appointment Management System offers several advantages over traditional appointment booking systems.

Advantages

- Online Appointment Booking
- Paperless Management
- Time Saving

- Easy Doctor Search
 - Secure Login
 - Better Record Management
 - User-Friendly Interface
 - Faster Appointment Scheduling
 - Reduced Human Errors
 - Efficient Administration
 - Digital Record Maintenance
 - Improved Healthcare Service
-

14. Limitations

The current version of the project has the following limitations:

- No Online Payment Facility
- No Video Consultation
- No Mobile Application
- No SMS Notifications
- No Email Notifications
- Single Hospital Support
- Internet Connection Required

These limitations can be addressed in future versions.

15. Future Scope

The Practo Doctor Appointment Management System can be enhanced by implementing additional features in future versions.

Future enhancements include:

- Online Payment Gateway
- Video Consultation
- AI-Based Doctor Recommendation

- Online Chat with Doctors
 - Mobile Application (Android & iOS)
 - Email Notifications
 - SMS Notifications
 - Laboratory Test Booking
 - Medicine Delivery
 - Ambulance Booking
 - Online Prescription Download
 - Multi-Hospital Support
 - Patient Feedback and Reviews
 - Health Report Management
 - Appointment Reminder System
-

16. Project Summary

The Practo – Doctor Appointment Management System is a modern web application developed using Python (Django Framework), HTML, CSS, JavaScript, and MySQL. The system provides an efficient platform where patients can register, search for doctors, and book appointments online without visiting hospitals physically.

The application consists of two primary roles: Admin and User. The Admin manages doctors, users, and appointments, while the User can register, log in, search doctors, view doctor details, and book appointments.

The project minimizes paperwork, improves healthcare management, ensures secure data storage, and provides a responsive user experience. The use of Django and MySQL ensures that the application is scalable, secure, and maintainable.

Overall, the system offers a reliable digital healthcare solution that simplifies appointment scheduling and improves the interaction between patients and healthcare providers.

17. Conclusion

The Practo – Doctor Appointment Management System successfully fulfills the objective of providing an efficient and user-friendly online appointment booking

platform. The system replaces the traditional manual appointment process with a secure, reliable, and digital solution.

The project helps reduce waiting time, improves appointment management, enhances data security, and simplifies hospital administration. By using Django as the backend framework and MySQL as the database, the application provides a robust and scalable architecture suitable for future enhancements.

The project demonstrates how modern web technologies can improve healthcare services and deliver better experiences to both patients and administrators.
